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Semi-empirical lumped models of polymer pyrolysis for poly(methyl methacrylate) and polyoxymethylene

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A B S T R A C T

Quantitative lumped-kinetics models are constructed for pyrolysis of poly(methyl methacrylate) (PMMA) and polyoxymethylene (POM) by using known reaction classes to describe mass loss (volatiles loss) by purely 1st-order decomposition rates, averting the need for molar- or number-based concentrations. These semi-empirical models will aid in establishing fundamental kinetics of polymer decomposition for solid rocket fuels and thermal recycling. Simultaneous thermogravimetric analysis and differential scanning calorimetry (TGA-DSC) are applied to measure mass-loss and heat-consumption rates and the influences of heating rate, sample size, and end groups as a basis for modeling. This combination of rate data and products is useful for proposing pathways and establishing mechanisms. PMMA and POM homopolymers were selected for base-case pyrolysis studies due to their relatively simple structures and their tendencies to yield primarily monomer. For comparison, kinetics was also measured for POM copolymer, where -CH_2CH_2O- units are interspersed among the -CH_2O- units.

Two-, three-, and four-lumped-reaction parameterized models are presented for pyrolysis rates and yields from POM homopolymer, POM copolymer, and PMMA, respectively. The lumped reactions correspond to temperature regions that are dominated by a single type of first-order reaction, each with a mass fractional yield of volatiles, an Arrhenius pre-exponential factor, and an activation energy. The first, lowest-temperature lump may be pericyclic reactions to molecular intermediates, or the main chain or weakly bound end groups or side groups may homolytically scission. Polymer-radical fragments could be trapped by recombination and be too large to be volatile. If enough polymeric radicals are formed, beta-scission into monomers can be rate-limiting for volatiles formation, and at higher temperatures, homolytic scission would be rate-limiting. At highest temperatures, stages can be rate-limited by internal H-abstraction or termination via exothermic reactions to make strongly bound char residues.

1. Introduction

Quantitative models are key to creating next-generation propellants, plastic recycling processes, and understanding of pyrolysis fundamentals. The current understanding of polymer degradation varies widely from polymer to polymer. As a result, it is important to study a breadth of polymers that have diversity in structure, mechanisms, and pyrolysis products. In this work, two polymers are studied, polyoxymethylene (POM) and poly(methyl methacrylate) (PMMA), due to their tendencies to produce primarily monomer as pyrolysis products.

Previous studies on POM have focused on plastic recycling. As a result, many POM studies have focused on products of decomposition rather than the rate or mechanism of decomposition [1]. Mucha first identified pyrolysis of POM homopolymer as being initiated via random main-chain scission, followed by unzipping [2]. Once 160 °C is reached, an autoxidative scission reaction begins by formaldehyde accelerating degradation [3]. Archodoulaki and Seidler found that POM copolymers are more resistant to hot temperatures than POM homopolymer [4]. Several studies have identified mostly formaldehyde and some carbon dioxide as products of decomposition [2,4,5].

PMMA pyrolysis has been studied for many decades. Numerous studies have identified competing elementary reactions depending on reactor conditions [6–10]. However, kinetic models are often only reported as an apparent, overall single-stage degradation mechanism based on experimental thermogravimetric analysis (TGA) data. Kashiwagi et al. [7] found that thermal decomposition of anionically

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polymerized PMMA was only by random scission, while free-radical-polymerized PMMA was initiated by three homolytic-scission reactions of increasing bond energies: monomer head-to-head linkages, vinylidene-containing end groups, and random chain scission. Other notable detailed studies were by Hu and Chen [11] and Arisawa and Brill [12].

Direct integration of detailed thermochemical polymerization or polymer-pyrolysis mechanisms must include very many species. The method of moments gives composite results [13,14] and coupled matrix-based Monte Carlo (CMMC) [15] and kinetic Monte Carlo (kMC) [16,17] methods have also been applied effectively. Stoliarov et al. developed a mechanism-free Reactive Molecular Dynamics modeling approach, which revealed how PMMA physical interchain interactions affected initiation kinetics through cage effects and mechanical ratcheting apart of bonds [18].

In this work, kinetic parameters are estimated by using thermogravimetric analysis with differential scanning calorimetry (TGA-DSC) data, fitted to semi-empirical lumped-species models. The provided kinetic information serves to help fill the gap for simulations requiring a greater amount of detail than a single-stage overall approach but less than that of a full degradation simulation. The kinetic information in these models can then be applied to plastic recycling, pyrolysis, and combustion models.

2. Methods and experimental

2.1. Materials

Experimental measurements used commercially sourced POM homopolymer (Delrin®) and POM copolymer, both supplied by McMaster-Carr, and PMMA from Millipore Sigma. No further characterization was available from the suppliers.

POM homopolymer contains a repeating -O-CH_2- backbone; for POM copolymer, ~1–2 % (industry standard) [19] of the oxymethylene units are replaced by oxyethylene units (-O–CH_2CH_2-). Thermal stability of commercial POM polymers is due to their esterified end groups [20].

PMMA has a carbon backbone with methyl and methacrylate side groups. The end groups for the PMMA sample are unidentified but may be vinyl, methacrylate, or vinylidene-containing. Previous literature has identified that apparent activation energies are influenced by end-group chemistry [11].

2.2. TGA-DSC

Simultaneous thermogravimetric analysis with differential scanning calorimetry (TGA-DSC) was performed using a TA Instruments SDT Q600 unit. To remove moisture content from the polymer, samples were held overnight in an oven at 105 °C. Samples were then placed in alumina pans and subjected to heating from 100 °C to 550 °C at various heating rates. A sample size of 5 mg was selected for experiments exploring the effects of heating rate. Heating rates varied from 5 to 40 °C/min with a focus on 5, 10, 20, and 30 °C/min. TGA-DSC experiments were performed under a continuous flow of nitrogen (N_2) at 100 mL/min. All experiments started by equilibrating the system at 100 °C, followed by applying a heating rate to the final temperatures of 500 °C or 550 °C. All experiments were performed in duplicate, at minimum. For the heating rates of 5,10,20 K/min, characteristic times to 1 %, 50 % and 99 % destruction were compared. The results can be examined in supplementary information.

2.3. Data analysis and kinetic models

To estimate kinetic parameters such as activation energies and pre-exponential factions, first-order kinetics is assumed with respect to mass, which initially is proportional to the number of monomer units present. Resulting reaction rate coefficients, k, then are fitted to Arrhenius temperature dependences:

$$\frac{dm}{dt} = k \cdot m = A e^{\frac{-E_a}{R_g T}} \cdot m \tag{1}$$

where m is the mass of the sample, dm/dt is the rate of mass loss due to volatilization with respect to time, A is the pre-exponential factor, E_a is the activation energy, T is the absolute temperature, and R_g is the gas constant. Linearizing the Arrhenius expression, E_a and A are determined from ln *(k)* vs. $1/T$. Activation energies can be estimated by this method, although they rely on the assumption that mass-loss kinetics are first-order in unreacted polymer. Using a single-reactant model solely treats the entire sample as being a homogeneous mass of unreacted polymer [19] gradually being converted to volatiles in steps.

Lumped Arrhenius models are developed here that incorporate molecular reaction classes. They work well here for POM and PMMA due to their degradations mainly to monomers, and they simplify application to CFD modeling. Using mass instead of the number of specific molecular fragments also better reflects the TGA data, where mass loss is measured directly.

Semi-empirical molecular framing allows a compromise between the computational expense of detailed kinetic models and capturing essential molecular behavior. An example of a three-stage lumped kinetic model is described below:

$$P_0 \rightarrow P_1 + V_1;\ k_1 \tag{2}$$

$$P_1 \rightarrow P_2 + V_2;\ k_2 \tag{3}$$

$$P_2 \rightarrow R + V_3;\ k_3 \tag{4}$$

where P_0 represents the initial polymer (e.g., PMMA), P_i is an intermediate polymer species, V_i is a volatile product, and R is residue or char.

At this level of modeling, products of decomposition of the two polymers are treated solely as CH_2O and methyl methacrylate, respectively, based on their dominance in our prior measurements [21]. Assuming monomer products is a reasonable approximation, as our measurements and the literature on PMMA decomposition report only small amounts of other species [12,21]. Within the lumped models, the starting polymer chains are assumed uniform.

Chains can first undergo homolytic scission, resulting in a fragment-end radical that can unzip monomers by β-scission. The potential initiation step may be end-group scission, random chain scission, or side group/branch scission. A two-stage model of decomposition is employed for POM homopolymer, while a three-stage model is used for copolymer. PMMA requires the use of a four-stage model to capture the degradation behavior fully.

We have constructed a series of ordinary differential equations (ODEs) using Arrhenius reaction kinetics and developed lumped models to compare to our TGA data. A set of ODEs for a three-stage lumped model with linear *T(t)* rise is:

$$\frac{dP_1}{dt} = (1 - x_0) \cdot A_0 e^{\frac{-E_0}{R_g \cdot (T_i + \beta \cdot t)}} \cdot P_0 - A_1 e^{\frac{-E_1}{R_g \cdot (T_i + \beta \cdot t)}} \cdot P_1 \tag{5}$$

$$\frac{dR}{dt} = (1 - x_2) \cdot A_2 e^{\frac{-E_2}{R_g \cdot (T_i + \beta \cdot t)}} \cdot P_2 \tag{6}$$

$$\frac{dV_1}{dt} = (x_0) \cdot A_0 e^{\frac{-E_0}{R_g \cdot (T_i + \beta \cdot t)}} \cdot P_0 \tag{7}$$

where x_i represents the mass-yield ratio from P_i, T_i is the initial temperature, and β is the heating rate. The adjustable parameters for each reaction are activation energy, pre-exponential factor, and volatilization percentage. The model predictions are then fitted to the experimental data.

3. Results and discussion

3.1. POM

Thermogravimetric analysis (TGA) demonstrates different degradation patterns between POM homopolymer and POM copolymer. TGA results are shown in Figs. 1 and 2 for the POM homopolymer and the POM copolymer at a heating rate of 20 °C/min and a sample mass of 5 mg.

In agreement with previous studies [4], decomposition of the homopolymer occurs at lower temperature than the copolymer due to the difference in the structures. The C—C σ bonds found in POM copolymer are somewhat stronger than C—O σ bonds, resulting in these bonds having greater thermal stability compared to the homopolymer. More energy is required for homolytic chain scission within the copolymer, leading to a higher onset degradation temperature than in the homopolymer. Furthermore, the copolymer has a faster degradation rate than the homopolymer, as illustrated in Fig. 2. The difference in degradation rate can also be explained by the difference in thermal stability of the homopolymer versus the copolymer. The C—C bonds in the copolymer stop β-scission reactions from proceeding down the backbone. However, in the homopolymer, the lack of C—C bonds allows β-scission to continue fully. Because the copolymer begins degradation at a higher temperature, there is more energy in the system to accelerate degradation. Another difference in the degradation patterns of the POM samples can be seen around 350 °C. The derivative curve in Fig. 2 clearly demonstrates a change in the degradation pattern of the copolymer that is not present in the homopolymer. The shoulder in the derivative curve of the copolymer signifies the point at which C—C bonds are being broken during random chain scission as opposed to the breaking of C—O bonds during homopolymer degradation.

Changes to heating rates were reflected in large changes to the degradation patterns, particularly for the homopolymer. TGA results are shown in Fig. 3 for POM homopolymer and copolymer at a heating rate of 30 °C/min and a sample mass of 5 mg.

With an increase of 10 °C/min to the heating rate, the degradation pattern of the homopolymer changes drastically, showing a shoulder on the mass-loss curve at around 350 °C. In contrast, the copolymer maintains a characteristic degradation pattern at both heating rates. Thus, we conclude that the behavior exhibited by the homopolymer is chemical in nature and is not an effect of heat-transfer issues sometimes seen in TGA experiments at high temperatures.

Figs. 4 and 5 further demonstrate the effect of heating rate on POM degradation. The shoulder in the homopolymer mass loss curve first appears at a heating rate of 30 °C/min and continues to appear at higher

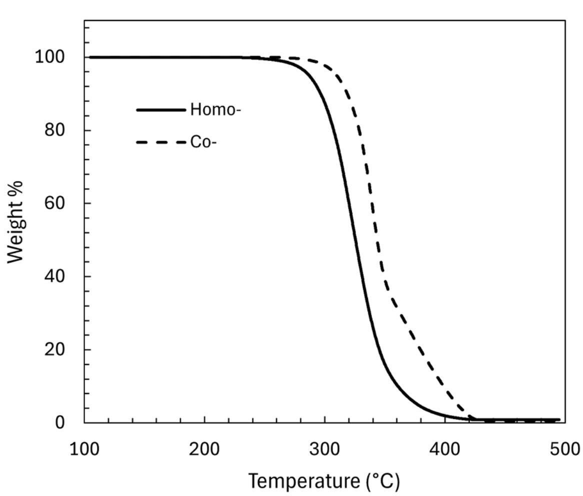

Fig. 1. Mass loss of POM homo- and copolymer at heating rate 20 °C/min.

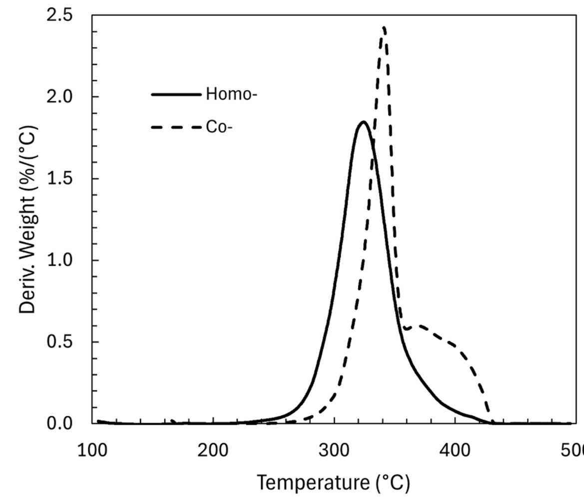

Fig. 2. Derivative mass loss of POM homo- and copolymer at heating rate 20 °C/min.

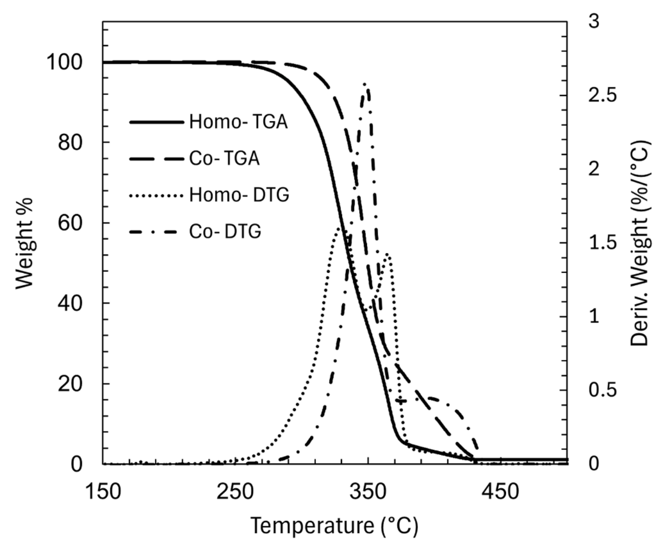

Fig. 3. TGA data of POM homo- and copolymer at heating rate 30 °C/min.

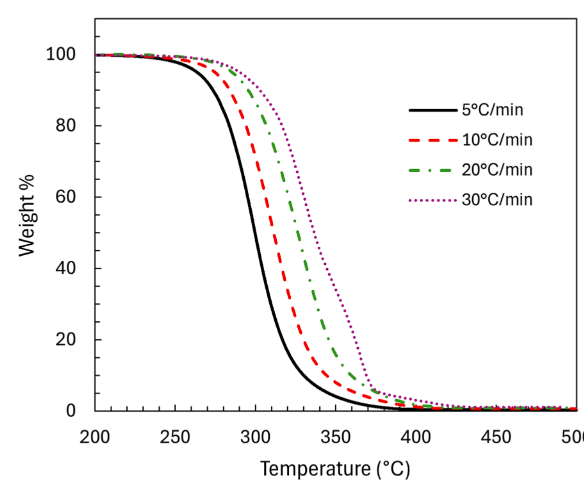

Fig. 4. Mass loss of POM homopolymer at various heating rates.

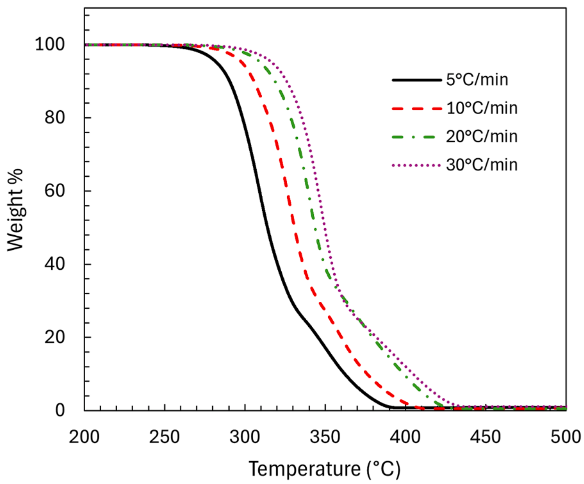

Fig. 5. Mass loss of POM copolymer at various heating rates.

heating rates. The shoulder may appear at higher heating rates because the unzipping reaction lags the initiation step. However, this phenomenon is not yet fully understood and should be studied further. Copolymer largely exhibits the same patterns at all heating rates.

Differential scanning calorimetry (DSC) data for POM homopolymer and copolymer are shown in Figs. 6 and 7 for a heating rate of 20 °C/min and a sample mass of 5 mg. DSC allows the detection of reactions and phase changes that do not produce volatile products and so do not appear on a mass loss curve. The melt temperature for the homopolymer is ~180 °C, while the copolymer is ~165 °C. Heat flow during degradation for the homopolymer largely mirrors the derivative mass-loss curve, implying one main reaction mechanism as the basis for degradation. Within the homopolymer DSC curve, there is a peak around 400 °C, possibly due to char formation. Comparatively the copolymer also displays one main reaction region but exhibits two smaller peaks around 400 °C. This observation is indicative of secondary reactions involving the remaining C—C bonds. These DSC results form the basis for modeling homopolymer as a two-reaction process and copolymer as a three-reaction process.

Examining the reaction regions carefully through the lens of TGA/DSC data allows determination of basic parameters using Arrhenius kinetics, as described previously. Kinetic parameters found using Arrhenius analysis serve as initial guesses for the lumped models before

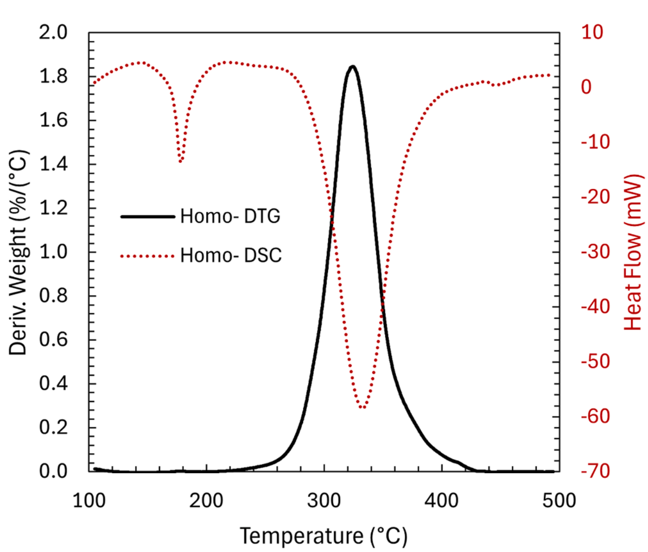

Fig. 6. TGA/DSC data of POM homopolymer at heating rate 20 °C/min.

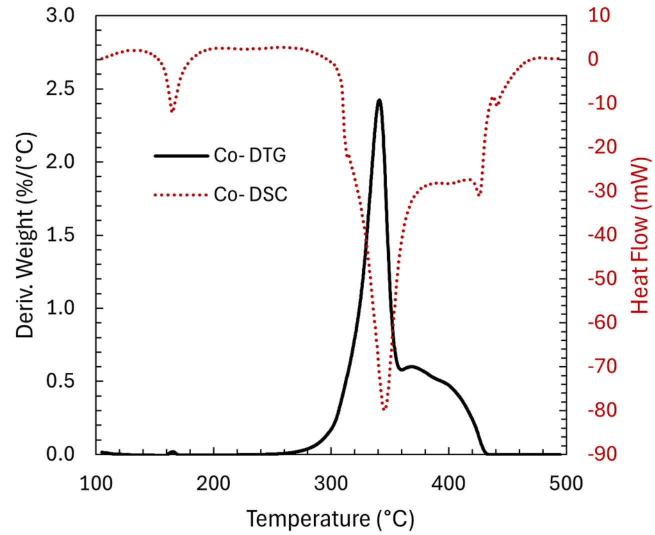

Fig. 7. TGA/DSC data of POM copolymer at heating rate 20 °C/min.

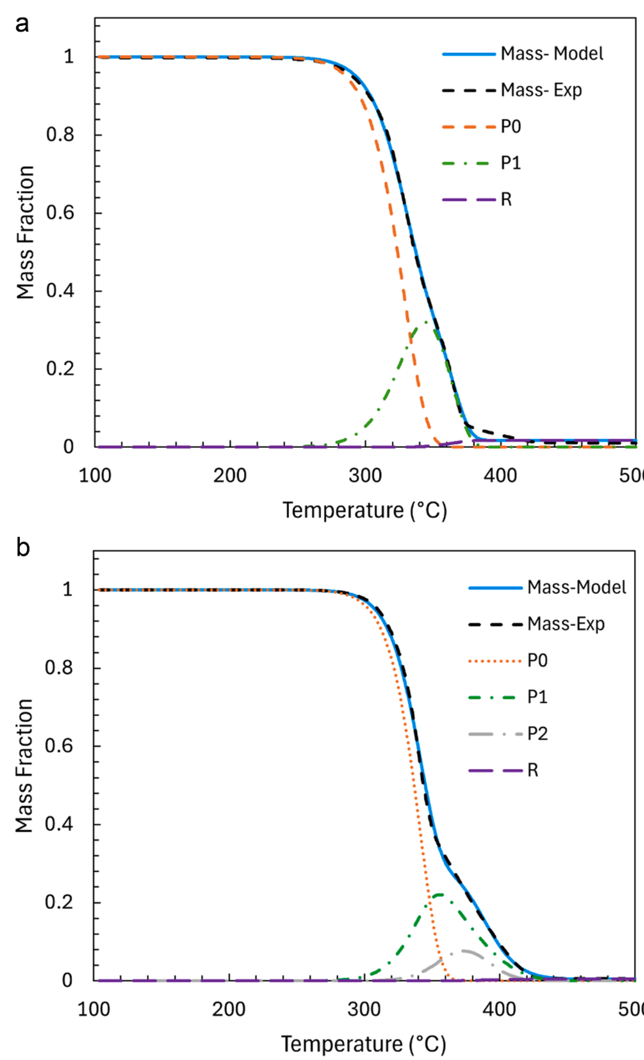

Fig. 8. (a) Two-stage lumped model of POM homopolymer at heating rate 10 °C/min; (b) Three-stage lumped model of POM copolymer at heating rate 10 °C/min.

kinetic variables are parameterized to fit experimental data. POM homopolymer is modeled as a two-stage lumped model, and POM copolymer as a three-stage model.

The two models' fits against experimental data are shown in Fig. 8, as well as the evolution of intermediate species as the degradation progresses. Homopolymer degradation is almost entirely dominated by the first stage, with the smaller second stage accounting for some char formation. Based on previous FTIR data [22], the mechanism for initiation is likely main-chain homolytic scission. However, this initiation step will not produce any volatiles directly and thus does not appear as a TGA curve. The first stage of the proposed model is then assumed to be β-scission down the polymer backbone. Neither the lumped model nor the TGA/DSC data detects an end-group-scission initiation step.

The copolymer lumped model is presented as having three stages due to presence of oxyethylene units. Stage one of the homopolymer and copolymer models may be β-scission down the POM backbone.

The added stage to POM copolymer model may represent a reaction occurring at a C—C bond, present only in the copolymer. The copolymer lumped model is presented as having three stages due to presence of oxyethylene units. Stage one of the homopolymer and copolymer models may be β-scission down the POM backbone.

The lumped models capture some differences among degradation patterns at different heating rates. Fig. 9 shows that the increase in degradation start temperature due to an increase in heating rate is modeled correctly for both polymers. The two-stage lumped model for homopolymer fails to capture the shoulder present at 30 °C/min, while the copolymer three-stage model retains the characteristic shoulder at all heating rates.

The quantitative models are presented in Table 1, fitting the mass-

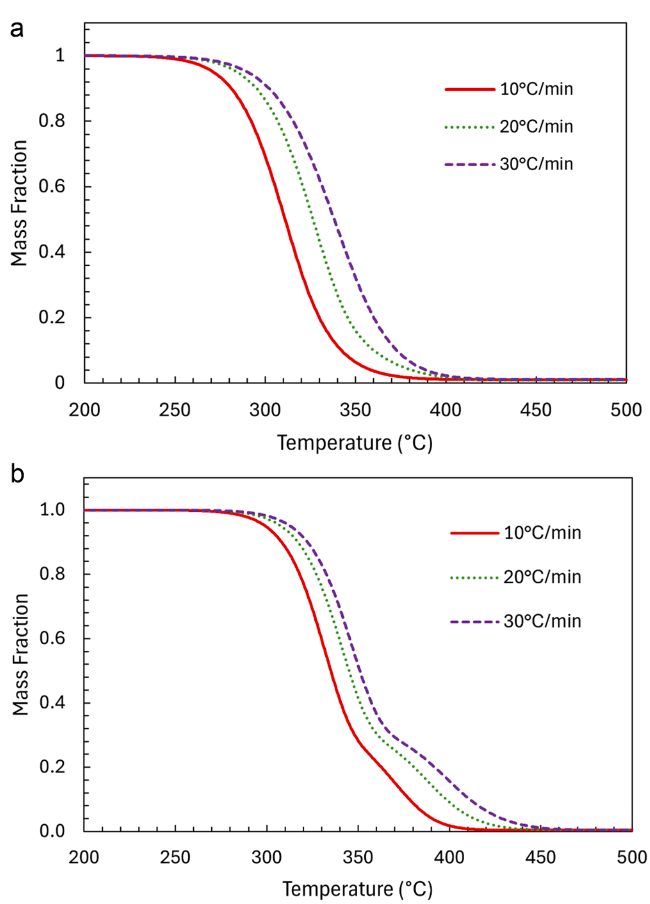

Fig. 9. (a) Two-stage lumped model of POM homopolymer at various heating rates; (b) Three-stage lumped model of POM copolymer at various heating rates.

Table 1
Average kinetic parameters for POM homopolymer and copolymer lumped models at 10 °C/min; mass basis for forming formaldehyde ($V_1 = V_2 = V_3 = CH_2O$) and carbon residue, *R*.

Reaction	Pre-exponential factor *A*, kg/(kg reactant s)	Activation energy E_a, kcal/mol	% mass to volatiles	Mass stoichiometric reaction
Homopolymer				
$P_0 \rightarrow P_1 + V_1$	$2.51 \times 10^{14} \pm 1.68 \times 10^{14}$	43.4 ± 0.986	63.0 ± 1.6	hPOM → 0.387 P_1 + 0.613 CH_2O
$P_1 \rightarrow R + V_2$	1.49 ± 0.75	6.40 ± 0.68	98.0 ± 0.09	P_1 → 0.020 *R* + 0.980 CH_2O
Copolymer				
$P_0 \rightarrow P_1 + V_1$	$6.24 \times 10^{16} \pm 3.6 \times 10^{14}$	51.3 ± 0.1	50.2 ± 6.8	cPOM → 0.498 P_1 + 0.502 CH_2O
$P_1 \rightarrow P_2 + V_2$	$2.65 \times 10^{4} \pm 1.5 \times 10^{1}$	19.1 ± 0.3	95.8 ± 6.0	P_1 → 0.042 P_2 + 0.958 CH_2O
$P_2 \rightarrow R + V_3$	$1.32 \times 10^{11} \pm 2.65 \times 10^{8}$	39.5 ± 0.2	21.9 ± 1.7	P_2 → 0.781 *R* + 0.219 CH_2O

loss curve well. The maximum deviation between the non-volatile mass of the POM homopolymer model and the experimental mass for any experiment, at the heating rate of 10 °C/min and a sample mass of 5 mg at any temperature is 1.9 %, occurring at 315.5 °C. The maximum deviation between the non-volatile mass of the POM copolymer model and the experimental mass for any experiment, at the heating rate of 10 °C/min and a sample mass of 5 mg at any temperature is 1.1 %, occurring at 355 °C. The stoichiometric coefficients are based on polymer mass, not molar concentration. Kinetics in polymers is not proportional to molar concentration of the polymer but rather depends on the number and types of reaction sites. Product volatiles are also shown on a mass basis. Because product volatiles have specific molecular weights, their yields are easily converted to molar production rates for use in gas-phase combustion kinetics.

3.2. PMMA

TGA analysis of PMMA reveals more complex degradation behavior than in POM homopolymer or copolymer. Fig. 10 depicts the mass loss curve and derivative mass-loss curve of PMMA at a heating rate of 20 °C/min and a sample mass of 5 mg. The derivative curve begins to increase slightly at about 180 °C, indicating an initial region of volatile loss,

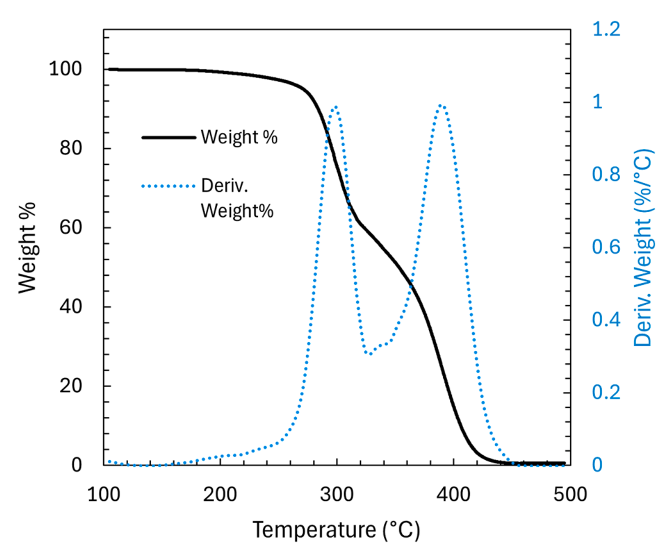

Fig. 10. Mass loss and derivative mass loss data for PMMA at heating rate 10 °C/min.

before rapidly increasing at 260 °C. Two distinct peaks can be seen in the derivative curve, occurring around 300 °C and 390 °C, representing two separate reactions. They may correspond to the end-group and random-scission reactions of Kashiwagi et al. [7].

The two separate PMMA reaction regions also appear in the DSC heat-flow data, shown in Fig. 11 along with the mass-loss curve. Heating rate is 20 °C/min, and sample size is 5 mg. A small initial change in the heat-flow curve is observed around 180 °C, corresponding to the melting point of PMMA.

The coupled rise in the derivative-mass curve and the corresponding endothermic peak in the heat-flow curve implies that PMMA decomposition begins immediately at the onset of melting.

Due to the small amount of mass loss, the initial step for this sample of PMMA may be end-group scission or degradation. This hypothesis assumes the sample has end-groups that are weakly bonded to the chain or that CO_2 is eliminated from them. Moens et al., found head-to-head linkages are the likely sites for initiation of depolymerization (the "head" carbon of a PMMA monomer is said to be the tertiary carbon along the backbone, bonded to the methyl and methacrylate groups) [15]. However, free-radical polymerization (FRP) of PMMA using azobisisobutyronitrile (AIBN) can result in vinyl end-groups via chain transfer termination [6,11]. Bond energies estimated via group additivity [23] show that vinyl end groups have a weak bond energy of 63.5 kcal/mol close to the estimated head-to-head bond energy of 64.8 kcal/mol. Moreover, end-group scission is less likely to participate in a recombination reaction due to an inverse cage effect [6], resulting in a small amount of volatile mass loss.

A small peak is in the heat-flow data at around 350 °C, corresponding to a possible exothermic reaction. At the end of the heat-flow data, around 460 °C, another exothermic peak may correspond to char formation.

Additional experiments show the degradation pattern of PMMA is unaffected by changes in initial sample mass or heating rate, in contrast to POM. Using TGA/DSC data to inform model creation, a four-stage model was created using the previously discussed method of calculating Arrhenius parameters as initial predictions. The lumped model, presented in Table 2, gives quantitative predictions that fit very well, as shown in Fig. 12. The maximum deviation between the non-volatile mass of the model and the experimental mass for any experiment at the heating rate of 10 °C/min and a sample mass of 5 mg at any temperature is 2.9 %, occurring at 395 °C.

The low activation energies reported for stages 1 and 3 suggest mass-transfer-limited release of volatiles. However, the notion of mass-transfer- limited reaction suggests that volatiles escape from the melt

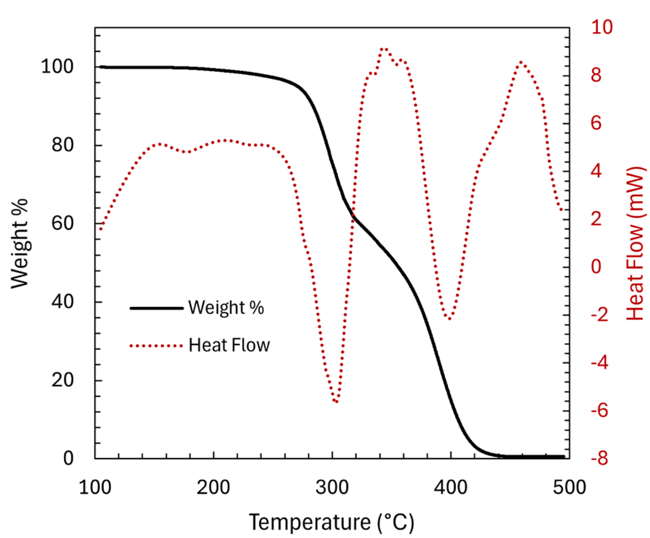

Fig. 11. Mass loss and DSC data for PMMA at heating rate 10 °C/min.

Table 2
Average kinetic parameters for PMMA lumped model at 10 °C/min, mass basis for forming methyl methacrylate ($V_1 = V_2 = V_3 = V_4$ = MMA) and carbon residue R.

Reaction	Pre-exponential factor A, kg/(kg reactant s)	Activation energy E_a, kcal/mol	% mass to volatiles	Mass stoichiometric reaction
$P_0 \rightarrow P_1 + V_1$	$6.97 \times 10^2 \pm 1.75 \times 10^1$	13.1 ± 0.05	4.9 ± 0.1	PMMA → 0.951 P_1 + 0.049 MMA
$P_1 \rightarrow P_2 + V_2$	$1.08 \times 10^{14} \pm 9.16 \times 10^{11}$	39.1 ± 0.59	35.8 ± 0.4	$P_1 \rightarrow 0.642\ P_2$ + 0.358 MMA
$P_2 \rightarrow P_3 + V_3$	84.9 ± 0.1	11.7 ± 0.16	19. ± 0.1	$P_2 \rightarrow 0.81\ P_3$ + 0.19 MMA
$P_3 \rightarrow R + V_4$	$7.99 \times 10^{11} \pm 9.05 \times 10^7$	42.4 ± 0.07	98.7 ± 0.6	$P_3 \rightarrow 0.013\ R$ + 0.987 MMA

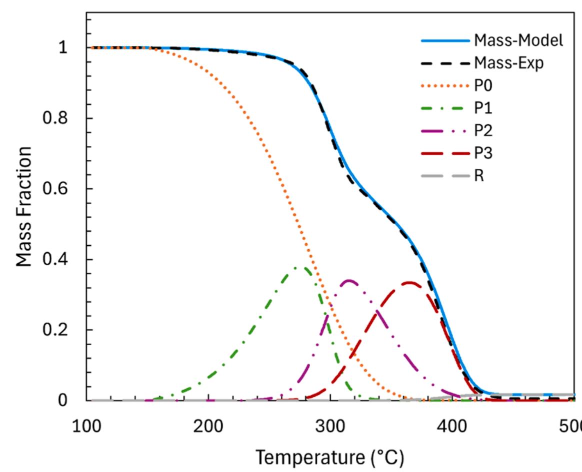

Fig. 12. Four-stage lumped model of PMMA at heating rate 20 °C/min.

layer and should be studied further to understand the phenomena at hand. Stages 2 and 4 have activation energies suggesting homolysis and/or beta-scission reactions [24].

The activation energies for stages 2 and 4 are in good agreement with overall estimations found by Korobeinichev et al. [10]. Furthermore, stages 2 and 4 have pre-exponential factors and activation energies similar to scission reactions found by Moens et al. [15]. Stage 3 is also in good agreement with β-scission reactions found by Moens et al. [15].

4. Conclusions

The results of this work are semi-empirical lumped kinetic models for the decomposition of POM and PMMA that reflect molecularly plausible reaction types.

The kinetic approach used in this work will be applied to different polymers such as polypropylene, polyethylene and hydroxyl-terminated polybutadiene, along with truly molecular models. Furthermore, volatile products of pyrolysis will be measured using gas chromatography-mass spectrometry (GC–MS). GC–MS data will be used to refine the lumped volatiles within the model into specific species and to connect semi-empirical models to detailed kinetic models.

Novelty and significance statement

The novelty of this research is the development of quantitative, semi-empirical lumped-kinetic models for polymer pyrolysis that bridge detailed reaction mechanisms with experimental thermal-degradation data. It is significant because framework is provided for the exploration of more complex polymers in the future. This work applies molecularly based reactions in the lumped-kinetic models for the thermal decomposition of two model polymers, poly(methyl methacrylate) (PMMA) and polyoxymethylene (POM). These models balance accurately capturing molecular behavior and computational simplicity, making them suitable for integration into combustion and recycling simulations. Our results not only provide insight into degradation pathways and kinetics of representative polymers but also serve as foundational models for more complex polymer systems and applications in energetic materials and sustainable material management.

Author contributions

Tim Mallo (First Author): Designed research, carried out experiments, wrote the paper, made figures, created PMMA models.

Adam Dumas (Second Author): Designed research, carried out experiments, wrote the paper, made figures, formatted paper, created POM models.

Phillip Westmoreland (Corresponding Author): Oversaw research, created fitting method, provided feedback, edited paper, secured funding for research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.proci.2025.105807.

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